

It doesn't matter how great or optimised the algorithms, they're useless without the hardware. You can't make a program written for PowerPC run on an Intel machine or vice versa. While the hardware is also nothing but a piece of junk without the software, it is the hardware which forms a system's base, the bed on which software sleeps. Interestingly, the word 'embedded' is built around the word 'bed'.

The importance of hardware increases manifold when it comes to embedded systems, because the software has to be written in congruence with the hardware. If you diluted the concept of embedded systems to its bare bones, you would have Apple's style of producing devices – binding the hardware close enough to the software so that both of them work in harmony ensuring the number of problems that would occur stays minimal. Embedded systems, however, have the hardware-software binding at a much lower level, and the hardware-software harmony is not a luxury but one of the core requirements of embedded systems.

To understand the hardware design, let's look at the different parts (call them components, if you will) of embedded systems' hardware. Let's start with the heart and soul – the processor.

Processor

Not just in the embedded systems, in other aspects of computing also the processor is said to be the heart of the entire system – without it, it's guaranteed that nothing will work. Choosing a processor in case of embedded systems is one of the most crucial decisions since it decides almost all other aspects of the system.

Embedded systems are built to handle specific tasks; sometimes, very specific tasks. Due to this dedicated-to-one-task approach, the processor gains all the more importance. You can find a range of processors in embedded systems, which can range from a separate microprocessor to one made as a part of the main board itself.

Separate Processors

You'd be wrong if you thought that the likes of Core i7 or Phenom could be used in an embedded system. But the fact doesn't disqualify other types of processors which can be bought separately as well. In most cases, you



Processors in embedded systems are designed for specific uses